Mohammad Amin Samadi

Researcher: Learning Sciences, Human–AI Interaction & Learning Analytics

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Summary

Mixed-methods researcher studying how people learn and build capability with AI. PhD candidate in Education (HCI / Learning Analytics) at UC Irvine; dissertation on human–AI collaboration in teams. I build the instruments and platforms my research questions require: I designed and lead TRAIL, a controlled-study platform for human–AI teaming with randomized assignment, survey instrumentation, and automated data capture. I also ship production LLM systems (currently a Claude-based agent with full evaluation infrastructure at ROI-AI One).

Education

Ph.D. Candidate, Education (HCI / Learning Analytics) **2021 – Present**
University of California, Irvine. Advisor: Nia Nixon *Irvine, CA*

- Dissertation (in progress): human–AI collaboration in teams, built on the TRAIL platform.
- UCI School of Education Vandell Dissertation Research Award; Best Short Paper Award, EDM 2025.

B.Sc., Computer Engineering **2015 – 2019**
Isfahan University of Technology. Advisor: Mohammad Hossein Manshaei *Isfahan, Iran*

Industry Experience

AI Scientist Intern **Jan 2026 – Present**
ROI-AI One (roiaione.com) – agentic systems for retail chargeback management *San Francisco, CA*

- Took the Root Cause Analysis tool from concept to production: a multi-agent workflow that ingests case documents, tracks findings in a live ledger, and determines why a chargeback occurred against industry standards; demoed by the CEO to customers.
- Built an observability and eval system for agentic workflows: visualizes tool calls, thinking blocks, and agent dispatch, and scores agent performance across multiple methods to auto-flag regressions before production.
- Built an email orchestration system that routes inbound messages to the correct workflow, then drafts and sends the outbound response, with every automated action gated behind a per-customer, per-action switch that ships off by default.
- Built document intake and portal automation: OCR-based ingestion (AWS Textract + LLM tool-use) paired with automation that logs into retailer portals, imports documents, and submits response packets end-to-end across five retailer portals.
- Built an auto-improvement backend that turns internal analyst feedback into a suggested fix, opened automatically as a pull request.
- Built the pipeline that turns production agent traces into GRPO/SFT training data (context-window management, Hermes-formatted tool-call data) to distill the agent's reasoning into a smaller, deployable model (Qwen); ran initial training tests to validate the pipeline.
- Migrated the team's instance infrastructure from a single Lightsail instance to EC2.

Research Experience

TRAIL: Team Research and AI Integration Lab **May 2023 – Present**
Lead researcher and engineer, Nixon Lab, UC Irvine *Irvine, CA*

- Designed and lead TRAIL, a research platform that embeds configurable LLM teammates into real-time, multi-party chatrooms for controlled studies of human–AI collaboration; deployed on AWS and used across multiple lab studies with 50+ concurrent participants.
- Built the measurement toolkit the lab's studies run on: a self-hosted survey module, an A/B framework with randomized team assignment and automated data capture, RAG-grounded study materials, and collaborative experiment sharing with role-based permissions and version history.
- Led qualitative coding of team discourse data: developed coding schemes, trained and managed coder teams, and established inter-rater reliability before analysis.

- Designed and administered survey instruments for studies of team dynamics and perceptions of AI teammates.
- Analyzed team interaction data with mixed-effects models, ANOVA, and psycholinguistic methods (LIWC, discourse analysis), linking team discourse dynamics to collaboration outcomes; complemented statistical findings with qualitative analysis of chat transcripts.
- Publish public-facing research writeups of TRAIL studies at aminsamadi.org/blog.
- Architected the AI teammate as a multi-agent pipeline (hierarchical memory, Big-Five persona compiler, turn-taking rules calibrated against real chat data) with a simulation/replay tool for testing behavior changes against past experiments before deployment.
- Managed a cross-disciplinary team of research interns and MSWE capstone developers.

Multimodal Classroom Video Understanding Pipeline **Jan 2025 – Dec 2025**
github.com/aminsmd/multimodal-transcription, UC Irvine *Irvine, CA*

- Built an AI pipeline that transcribes and semantically segments classroom video to support research on teacher–student interaction quality; multimodal (audio + visual) transcription via Gemini 2.5 Pro, segmentation via temporal embedding similarity, deployed at scale on AWS ECS.
- Led the design of a qualitative coding framework for teacher actions, then operationalized it in a multi-agent system that codes classroom video automatically.
- Built a human-coded evaluation set from the framework and used it to validate the multi-agent system's coding against expert judgments. Work published in two 2026 papers (see Selected Publications).

Skills

Research methods: mixed-methods research, experimental design, A/B testing, survey design, qualitative coding and inter-rater reliability, instrument development and validation, statistical modeling (mixed-effects models, ANOVA), psycholinguistic analysis (LIWC), discourse analysis, learning analytics, LLM evals and benchmark construction, UX research (HCI)

ML & LLMs: AI agents and multi-agent systems, Claude Agent SDK workflows, tool use / function calling, RAG, memory systems, prompt engineering, prompt caching, post-training (SFT, GRPO), reward modeling, LLM observability, multimodal LLMs (Claude, Gemini, OpenAI, Qwen)

Engineering: Python, R, TypeScript, SQL; PyTorch, Hugging Face Transformers, TRL, LangChain, CrewAI; Django, Flask, FastAPI, Next.js/React, Socket.IO; AWS (ECS, Fargate, S3, Textract), Docker, Terraform, PostgreSQL, GitHub Actions

Selected Publications

- J. Choi, **M. A. Samadi**, S. Jaquay, S. Park, N. Nixon. "Read the Room or Lead the Room: Understanding Socio-Cognitive Dynamics in Human-AI Teaming." *Learning Analytics and Knowledge (LAK '26).*
- **M. A. Samadi**, N. Nixon. "Personalities at Play: Probing Alignment in AI Teammates." Preprint, 2026.
- J. Wu, L. Li, **M. A. Samadi**, et al. "Validating High-Inference Pedagogical Constructs: Preliminary Evidence from a Multi-agent Architecture for Equity-Focused Mathematics." *International Conference on Artificial Intelligence in Education (AIED '26).*
- L. Li, **M. A. Samadi**, J. Wu, et al. "Automated Multimodal Transcription for Belonging-Centered Classroom Interaction Analysis: Opportunities and Challenges." *International Conference on Artificial Intelligence in Education (AIED '26).*
- **M. A. Samadi**, S. JaQuay, J. Gu, N. Nixon. "The AI Collaborator: Bridging Human-AI Interaction in Educational and Professional Settings." Preprint, arxiv.org/abs/2405.10460.
- **M. A. Samadi**, S. JaQuay, Y. Lin, E. Tajik, S. Park, N. Nixon. "Minds and Machines Unite: Deciphering Team Social and Cognitive Dynamics with AI." *Learning Analytics and Knowledge (LAK '24).*
- **M. A. Samadi**, D. Shariff, R. Ibarra, N. Nixon. "Balancing the Scales: Using GPT-4 for Robust Data Augmentation." *Companion Proceedings of LAK '25.*
- S. Park, D. Shariff, **M. A. Samadi**, N. Nixon, S. D'Mello. "From Discourse to Dynamics: Understanding Team Interactions Through Temporally Sensitive NLP." *Educational Data Mining (EDM '25).* **Best Short Paper Award.**
- K. Karbasi, K. Hong, **M. A. Samadi**, G. Pottie. "Multi-Agent Collaborative Framework for Math Problem Generation." *Educational Data Mining (EDM '25).*